

Tyler N. Morrison

Physical AI Research Engineer

913-944-2055 | tylormorrison7777@gmail.com | [tymo77](#) | [Google Scholar](#)

Experience

Boston Dynamics

Staff Research Engineer - Reinforcement Learning

Boston, MA

February 2026 — Present

- Trained and deployed **hierarchical RL** models for complex robotic manipulation tasks
- **Designed ML model export pipeline** for edge-device inference
- Contributed to team **research investigations in RL for dextrous manipulation**

Agility Robotics

Senior Robotics Software Engineer

Pittsburgh, PA and Remote

February 2022 — January 2026

- **Identified and solved novel technical problems in robotics, drove product-market fit, and scaled engineering for first five customer use cases.**
- **Led peers in a highly dynamic fast-paced start-up environment including company growth through multiple funding rounds.**
- Named **technical lead** of the Skill Development team of six which uses primitives to construct robot behaviors for use by Applications.
- **Technical lead** of a year-long critical inter-service communication project spanning from cloud to robot.
- Led two critical **world-first** commercial deployments ([link](#)) as **technical lead and interim manager**.
- **Technical lead and demo lead for over 140 robot-hours of continuous, live humanoid tradeshow demos** including the [MODEX 2024](#) ([link](#)) and [Promat 2025](#) live demos, and key individual contributor for [Promat 2023](#).
- Engaged with technical leads from controls and AI groups to determine requirements for business-critical technology development projects.
- Led international **high-impact demos** for engagements with press, high net worth individuals, Fortune 100 executives, investors, etc., resulting in 100's of millions of USD in raised capital, two strategic partnerships, and two pilot projects.
- **Product development:** Drove after-action product gap analyses for engineering leadership after deployments and customer engagements to ensure product-market fit.
- Developed software bug triage process which reduced the backlog of open bugs by over 50% over two years.
- **Humanoid robot behavior developer** for a dozen unique use-cases.
- Constructed our first whole-robot hardware-in-the-loop test pipeline for continuous integration.
- **Cross-functional developer:** Primary contributor for the cloud-to-robot interface for managing robot behavior commands. Key contributor to architecture definition phase of forthcoming generational hardware design revision.
- Architect and lead developer for CLI tool for joint workflow/behavior development and debugging in use by 20+ developers for years
- Integrated **object detection, manipulation, and navigation** systems into complex tote manipulation behaviors deployed with >99% task success and 0 major faults in 40 hours of runtime across ten robots.
- **T-shaped engineer:** Developed and shipped features for robotics software in areas throughout the stack including: **perception, manipulation, logging and diagnostics, cloud connectivity, hardware-in-the-loop testing**, and more...

Design, Innovation and Simulation Lab (DISL)

Graduate Research Assistant

The Ohio State University

August 2017 — January 2022

- Developed simulation tools for powered ankle prosthesis design using trajectory optimization for human gait adaptation prediction.
- Modeled, tested, and analyzed variable stiffness robotic arm links for use in physical human-robot interaction.
- Mentored high school, undergraduate, and MS student research projects.

NSF Interfaces and Surfaces REU

REU Research Assistant

Clemson University

May 2016 — August 2016

Education



The Ohio State University

Ph.D. Candidate in Mechanical Engineering

Columbus, OH, USA

Aug. 2017 — Dec. 2022

- Dissertation: *Computational Design Methods for Compliant Robotic Ankle Prostheses*

• Distinguished University Fellow & Dept. Award Recipient



The University of Tulsa

B.S. Mechanical Engineering

Tulsa, OK, USA

Aug. 2013 — May 2017

- Chapman Presidential Scholar

Skills and Previous Experience

Software	Bazel, Python, JavaScript, C/C++, Docker, Bash, Rust, HTTP, AWS, MQTT
Robotics	Behavior Trees, Object Detection [†] , Factor Graphs [†] , Kalman-filters [†] , IK, RL control [†] , Teleop, Demonstration learning
Modeling & Design	Blender, Solidworks, AutoCAD, Figma, Adobe Illustrator
Simulation	Mujoco, OpenSim, ANSYS, ABAQUS
Hardware	Arduino, Raspberry Pi, DC Motors, Lidar, Binocular depth [†] , Basic Circuit Design, EtherCAT [†]
Soft	Technical leadership, Hiring, Interim management, High-pressure demos, Executive-level communication
Languages	English, Modern Greek, Spanish

[†]familiar with as a consumer

Selected Academic Publications

Tyler Morrison, Hai-Jun Su. *Stiffness Modeling of a Variable Stiffness Compliant Link*. Mechanism and Machine Theory. 2020. «Published». DOI: [10.1016/j.mechmachtheory.2020.104021](https://doi.org/10.1016/j.mechmachtheory.2020.104021)

Tyler Morrison, Richard Jagacinski, Jordan Petrov. *Drivers' Attention to Preview and Its Momentary Persistence*. IEEE Transactions on Human-Machine Systems. 2023. «Published». DOI: [10.1109/THMS.2023.3266152](https://doi.org/10.1109/THMS.2023.3266152)

Tyler Morrison, Chunhui Li, Xu Pei, Hai-Jun Su. *A Novel Rotating Beam Link for Variable Stiffness Robotics Arms*. IEEE International Conference on Robotics and Automation. *Montreal, CA*. May 2019. «Published». DOI: [10.1109/ICRA.2019.8793833](https://doi.org/10.1109/ICRA.2019.8793833)

Tyler Morrison, Hai-Jun Su. *Gait Prediction for Prosthesis Design Evaluation*. ASME IDETC-CIE. *St. Louis, MO, USA*. November 2022. «Published». DOI: [10.1115/DETC2022-89739](https://doi.org/10.1115/DETC2022-89739)

Tyler Morrison, Dylan Trainor, Hai-Jun Su. *Optimization of the Compliant Drive Mechanism for a Prosthetic Ankle*. ASME IDETC-CIE. *St. Louis, MO, USA*. August 2020. «Published». DOI: [10.1115/DETC2020-22442](https://doi.org/10.1115/DETC2020-22442)

Selected Honors & Awards

Graduate

2017	Distinguished University Fellow , The Ohio State University	<i>Columbus, OH, USA</i>
2017	Department Supplementary Fellowship Award , The Ohio State University	<i>Columbus, OH, USA</i>

Undergraduate

2017	College of Engineering and Natural Sciences Steven J. Bellovich Medal , The University of Tulsa	<i>Tulsa, OK, USA</i>
2017	Senior Project – Most Valuable Team Member , The University of Tulsa, Mech. Eng. Dept.	<i>Tulsa, OK, USA</i>
2016	Putnam Competition Team Member , The University of Tulsa, Math Dept.	<i>Tulsa, OK, USA</i>
2016	Nominee, National Barry Goldwater Scholarship , The University of Tulsa	<i>Tulsa, OK, USA</i>
2013	Presidential Scholar , The University of Tulsa	<i>Tulsa, OK, USA</i>

Other

2011	Eagle Scout , Boy Scouts of America	<i>Kansas, USA</i>
------	--	--------------------

Passions

- Long-distance running and cycle touring: Multiple centuries completed including 200 mile solo across Ohio.
- Dog training: Adopted and rehabilitating a beloved reactive dog.
- Language learning and cultural exchange: Individual study in modern Greek to advanced level.